15.7 Experimental and Theoretical Probabilities from Simulations

Lesson Introduction

Benchmarks	MA.7.MA.DP.2.3 Find the theoretical probability of an event related to a simple experiment. MA.7.MA.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.	Learning Target	• I can determine the probability of an event from a simulation and compare it to the theoretical probability of the event.

	Building On	Working Toward
Benchmark Coherence	N/A	MA.8.DP.2.2 Find the theoretical probability of an event related to a repeated experiment. MA.8.DP.2.3 Solve real-world problems involving probabilities related to single or repeated experiments, including making predictions based on theoretical probability.

Essential Questions	• How do you determine a simple experiment that could be used to simulate the probability of an event? • How does the probability from a simulation compare to the theoretical probability of the event?
Lesson Narrative	In this lesson, students expand their knowledge of theoretical and experimental probabilities through simulations. Students practice conducting simulations to determine experimental probabilities. To start the lesson, students determine different simple experiments that could be used to simulate a probability. Students then conduct those experiments and record the data. They then use the data and compare the theoretical and experimental probabilities.

Component Summary	15.7.1 Warm-Up (~5 min)	Visualize probabilities (SE pg. 432)	15.7.3 Your Turn! (15 min)	Practice finding experimental and theoretical probabilities (SE pg. 434)
	15.7.2 Collaboration (20 min)	Practice simulating probabilities (SE pg. 432)	15.7.4 Wrap-Up (~5 min)	Self-reflect on the learning targets from the unit (SE pg. 435)

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: • Experimental probability • Theoretical probability • Simple experiment	• 6-sided die • 12 equal-sized paper slips numbered 1-12 • Standard deck of 52 cards • 4-section spinners	• If any materials are not available, electronic die or simulators can be used instead.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>In 15.7.3 Your Turn!, students may only count the cards between 11 and 25, not including those numbers. Consider prompting students to write out the numbers and count how many cards there would be.
Students may confuse theoretical and experimental probability.</td><td>Consider providing equations for solving for theoretical probability and posting them on the board for students to reference.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
Consider implementing a Think-Pair-Share for question 2 in 15.7.2 Collaboration. Students can work with their partner to come up with the different simple experiments that could be used to simulate Jimmy’s three-point shot probability.
As a class, partners can share what they came up with and discuss how each probability should be equivalent to $\frac{1}{4}$.</td><td>MA.K12.MTR.7.1: Real-World Contexts
Students may have experience with seeing probabilities used in the real world, for example, in sports. Use 15.7.2 Collaboration and conversation about statistics in sports to bridge probability and statistics, which are used in a variety of industries in the “real world.”</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">In 15.7.2 Collaboration, consider scaffolding by reminding students to goal is to create a fraction equivalent to $\frac{2}{3}$. You can provide students with the following fill in the blank equations to help…
For a six-sided die, $\frac{2}{3} = \frac{}{6}$
For a standard 52-card deck, $\frac{2}{3} = \frac{}{52}$</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

15.7.1 Warm-Up: Visualizing Probabilities

Component Overview: *Students visualize probabilities using double number line diagrams before interpreting a probability verbally.*

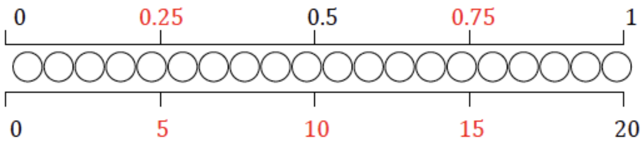
15.7.1 Warm-Up: Visualizing Probabilities

Ashley places 2 red marbles, 3 blue marbles, 4 yellow marbles, 5 orange marbles, and 6 purple marbles in a bag.

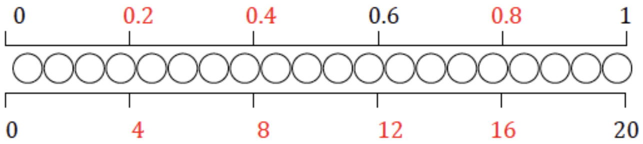
A representation of the marbles is shown.

1. Complete the double number line diagrams below by writing values at each tick mark. The top line represents probabilities, and the bottom line represents the number of marbles in the bag.

a.



b.



2. Write a color on each blank to complete the statements.

The probability of pulling a ______*yellow*______marble out of the bag without looking is 0.2. The probability of pulling an _____*orange*_____ marble out of the bag without looking is 0.25.

Prompt students to notice into how many sections the double number line is broken to determine the missing pieces.
If one whole is broken into 4 equal pieces, what is each piece worth?

15.7.2 Collaboration: Simulating Probabilities

Component Overview: *Students practice simulating probabilities by coming up with simulations using different simple experiments that could be used to describe Jimmy Butler's probability of making a three-point shot.*

15.7.2 Collaboration: Simulating Probabilities

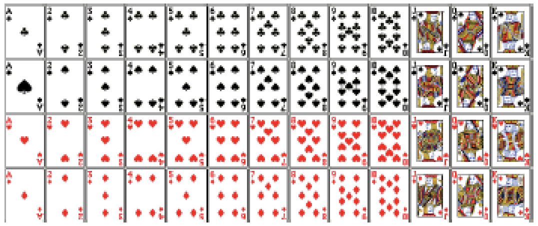
1. In the 2019-2020 NBA regular season, Miami Heat player Jimmy Butler made $\frac{1}{4}$ of the three-point shots he attempted. What was the probability of Jimmy Butler making a three-point shot in the 2019-2020 regular season, represented as a decimal?
   0.25

2. Work with your partner to complete each statement to describe ways Jimmy's three-point shot probability could be simulated using different simple experiments.

   a. Create a spinner with 4 equal section(s), where 1 section(s) represent(s) Jimmy Butler making a three-point shot and the other(s) represent(s) misses.

   b. Place twelve equal-sized slips of paper with the numbers 1 – 12 written on them in a bag. Papers with the numbers *answers will vary (any three numbers)* represent Jimmy Butler making a three-point shot and the others represent misses.

   c. Use a standard deck of 52 cards where *Answers will vary. (Any 13 cards)* represent Jimmy Butler making a three-point shot and the other cards represent misses.



Think-Pair-Share
Consider implementing a Think-Pair-Share for question 2. Students can work with their partner to come up with the different simple experiments that could be used to simulate Jimmy's three-point shot probability.
As a class, partners can share out what they came up with and discuss how each fraction should be equivalent to $\frac{1}{4}$.

Prompt students to use the visual of the cards to help.

15.7.2 Collaboration: Simulating Probabilities (continued)

3. Suppose in the first post-season game that year, Jimmy Butler attempted 5 three-point shots. Choose one of the simulations from the previous question and work with your partner or group to simulate the number of three-point shots Jimmy made.
Answers will vary.
   a. Repeat the simulation 5 times and record your results in the table.

	Shot 1	**Shot 2**	**Shot 3**	**Shot 4**	**Shot 5**	**Fraction Made**
Simulation 1						$\frac{}{5}$
Simulation 2						$\frac{}{5}$
Simulation 3						$\frac{}{5}$
Simulation 4						$\frac{}{5}$
Simulation 5						$\frac{}{5}$

   b. Based on your group's simulations, estimate the probability that Jimmy Butler made $\frac{2}{5}$ of the three-point shots he attempted in the game.
   Answers will vary.

   c. Jimmy Butler actually made 3 out of the 5 three-point shots he attempted in the game. Explain how that compares to your group's simulation results.
   Answers will vary.

Students will need to have spinners, papers, and standard decks of cards that they can use to perform the simulation.

If supplies are not available, a random number generator or virtual simulator can be used online.

Why is it that the experimental probabilities are not always the same as each other or to the theoretical probability?

4. In the 2019-2020 NBA regular season, Orlando Magic player Aaron Gordon made $\frac{2}{3}$ of free-throw shots attempted. Work with your partner to describe how each of the following could be used to simulate his attempted free-throw shots.

   a. A fair 6-sided die
   Answers will vary. Any four numbers on a six-sided dice will accurately simulate the number of free throws that Aaron Gordon will make.

   b. A standard 52-card deck
   Answers will vary. The number 52 is not divisible by three, so one card will need to be taken out of the deck. If one card is taken out, then any 34 cards in the deck would simulate the number of free throws that Aaron Gordon will make.

   c. Different colored marbles placed in a bag
   Answers will vary. Two-thirds of the colored marbles chosen would simulate the number of free throws that Aaron Gordon will make.

Consider scaffolding by reminding students the goal is to create simulations with probabilities equivalent to $\frac{2}{3}$. You can provide students with the following fill in the blank equations to help…

For a six-sided die, $\frac{2}{3} = \frac{\_\_}{6}$.

For a standard 52-card deck, $\frac{2}{3} = \frac{\_\_}{52}$.

15.7.3 Your Turn!

Component Overview: *Students continue conducting simulations to determine experimental probabilities. This time, students use a 6-sided die and record the data and use it to find different probabilities.*

15.7.3 Your Turn!

1. Suppose a fair six-sided die numbered 1 through 6 is rolled 12 times.

   a. Of those 12 rolls, how many do you expect will land on the number 3?

   2 times out of 12 rolls

   b. Roll a standard six-sided die 12 times and record the number on the top of the cube when it lands.

Roll 1	Roll 2	Roll 3	Roll 4	Roll 5	Roll 6	Roll 7	Roll 8	Roll 9	Roll 10	Roll 11	Roll 12

   c. What is the theoretical probability of rolling a 3?

   $\frac{1}{6}$

   d. Describe how the theoretical probability compares to your experimental probability.

   Answers will vary.

   e. Review your results and your partner's results. Explain why the results are not the same.

   Answers will vary. The more times that the dice is rolled, the closer the experimental probability will get to the theoretical.

Students will either need a die or access to a digital die.

Consider a reminder of how to find theoretical probability.

$$\frac{\textit{number of desired outcomes}}{\textit{total number of possible outcomes}}$$

Consider posting this formula on the board for students to reference.

Does the number of trials affect how close the theoretical and experimental probabilities are?

15.7.3 Your Turn! (continued)

2. Consider the events of rolling a fair 10-sided die and getting an odd number or flipping a coin and having it land tails-up. Explain which event is more likely.
   The events are equally likely because the theoretical probability of tossing a coin and it landing tails-up is $\frac{1}{2}$. The theoretical probability of rolling a ten-sided dice and landing on an odd number is $\frac{5}{10}$ that reduces to $\frac{1}{2}$.

3. Noelle picks a card at random from a stack that has cards numbered 11 through 25.

   a. Determine the sample space.
   11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25
   The sample space is fifteen cards.

   b. Find each probability.

   $$P(\text{even}) = \frac{7}{15} \text{ or } \approx 47\%$$

   $$P(\text{multiple of } 5) = \frac{3}{15} \text{ or } 20\%$$

   $$P(\text{two digit number}) = \frac{15}{15} \text{ or } 100\%$$

   $$P(\text{number less than } 10) = \frac{0}{15} \text{ or } 0\%$$

   c. Noelle used a simulator to run the experiment 300 times. She selected an odd-numbered card 154 times. Compare the experimental probability to the theoretical probability of selecting an odd-numbered card.
   The experimental probability is $\frac{154}{300}$ or 51.3%. The theoretical probability of picking an odd number is $\frac{8}{15}$ or 53.3%.

Students may only count the cards between 11 and 25, not including those numbers. Consider prompting students to write out the numbers and count how many cards there would be.

To extend students' thinking, have them research a sports player's statistics of their choosing.
Students can then come up with a simple experiment that could be used to simulate the probability researched.

15.7.4 Wrap-Up: Reflection

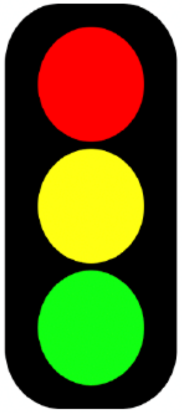
Component Overview: *Students reflect on their understanding of concepts addressed throughout the unit.*

15.7.4 Wrap-Up: Reflection

In this unit, you have learned multiple concepts about probability, including those listed below.

- Determine the sample space for a simple experiment
- Express probabilities as fractions, decimals, and percentages
- Interpret and compare the probability of chance events
- Find the theoretical probability of an event related to a simple experiment
- Compare experimental probability from simulations to theoretical probability

1. Use the stoplight reflection below to share how you are feeling about each of these concepts at this point.



I still feel stuck on how to ...
Answers will vary.

I'm starting to get the hang of ...
Answers will vary.

I feel confident and ready to go with ...
Answers will vary.

For an alternative approach to the Wrap-Up, have students use red, yellow, and green highlighters to highlight the concepts listed based on how they feel about each one.